FOR THE TUTOR OR PARENT

# Answer key: Ages 9-11

Accept equivalent forms (for example 2.05 and $2.05). One mark per question. Keep this key away from the child until they have finished.

## Warm-up (4 marks)

| # | Question | Answer | Watch for |
|---|---|---|---|
| 1 | A bag has 9 sweets. How many sweets are in 7 bags? | **63** | Check recall of 9 x 7; note if the child skip-counts slowly. |
| 2 | Fill in the blank: 2/3 = ?/12 | **8** | Does the child multiply top and bottom by 4? Common error: 2 + 9 = 11. |
| 3 | A rectangle is 7 cm long and 6 cm wide. What is its area? Write your answer with units. | **42 square cm** | Accept 42 square cm, 42 sq cm or 42 cm2. If the unit is missing, give the mark and note it. Common error: giving the perimeter, 26 cm. |
| 4 | 5000 - 1346 = ? | **3654** | Watch for borrowing across zeros; common error 4346 or 3764. |

## Main questions (6 marks)

| # | Question | Answer | Watch for |
|---|---|---|---|
| 5 | A factor of 18 is a number that divides 18 exactly with nothing left over. Write down all the factors of 18. | **1, 2, 3, 6, 9, 18** | Look for pairs (1 x 18, 2 x 9, 3 x 6). Common errors: missing 1 or 18, or listing multiples (18, 36...). |
| 6 | Round 48,572 to the nearest thousand. | **49,000** | Check which digit they look at (the hundreds digit, 5). Common error: 48,000 or 48,600. |
| 7 | 12 + 6 x 3 - 4 = ? | **26** | Multiplication first: 6 x 3 = 18, then 12 + 18 - 4. Common error: 50 (left to right). |
| 8 | 2 and 3/5 + 1 and 4/5 = ? Write your answer as a mixed number. | **4 and 2/5** | Mixed number preferred: 4 2/5. Accept 22/5 for the mark. Look for 3/5 + 4/5 = 7/5 = 1 2/5, then add the whole numbers. Common errors: 3 7/5 left unconverted, or 4 7/10. |
| 9 | A ribbon is 3.7 m long. Another ribbon is 2.85 m long. What is the total length? Write your answer in metres. | **6.55 m** | Check place-value alignment of the decimal points. Common errors: 3.22 (37 + 285 with the decimal point ignored) or 5.92 (digits lined up from the right). |
| 10 | A rectangle has a perimeter of 36 cm. It is 11 cm long. How wide is it? | **7 cm** | Look for 36 / 2 = 18, then 18 - 11 = 7. Common errors: 25 (36 - 11) or 14. |

## Brain stretchers (2 marks, optional)

| # | Question | Answer | Watch for |
|---|---|---|---|
| 11 | (Brain stretchers are hard on purpose. Most children do not finish both, and that is fine. Try one!) I am a number between 30 and 50. I am a multiple of 4 and a multiple of 6. My digits add up to 9. What number am I? | **36** | Look for listing multiples of 4 and 6 (common: 36, 48) then using the digit clue. Accept guess and check with reasoning. |
| 12 | Ben has 3 times as many marbles as Cara. Ben gives 12 marbles to Cara. Now they have the same number. How many marbles did Ben have at first? | **36** | Bar model: after the gift, Ben's 3 units minus 12 equals Cara's 1 unit plus 12, so 2 units = 24, 1 unit = 12. Cara 12, Ben 36. |

Questions the child ticked "not learnt yet" score 0 but tell you what to teach first. They do not count against the child.